

CKP
b. 01

2. $\bar{x} = 80$ $s = 20$ $n = 100$

95% : $80 \pm 1.96 \cdot \frac{20}{\sqrt{100}}$
97% : $80 \pm 2.17 \cdot \frac{20}{\sqrt{100}}$

1. $\bar{x} = 34$ $s = 12$

a) $P(28 < X < 31)$

$= \left(\frac{31-34}{12} \right) - \left(\frac{28-34}{12} \right)$

$= \Phi(-0.25) - \Phi(-0.5)$

$= 1 - \Phi(0.25) - (1 - \Phi(0.5))$

$= 1 - (0.5987) - (1 - 0.6915)$

$= 0.0934$

b) $P(X > 30) = 1 - P(X \leq 30)$

$= 1 - \left(\frac{30-34}{12} \right) = 1 - \Phi(-0.33)$

$= 1 - (1 - 0.6293) = 0.6293$

c) Median = per 50 = $\Phi(0.00) = 0.5$

$0.00 = \frac{X-34}{12} \Rightarrow X-34 = 0$

$\therefore X = 34$

C.L.T d) $n = 60 \Rightarrow P(X \leq 33) = P\left(Z \leq \frac{33-34}{12/\sqrt{60}}\right)$

$= 1 - \Phi(0.65)$

$= 1 - 0.7421 = 0.2579$

$H_0: \geq 100$ $H_1: < 100$

2. $\bar{x} = 91.4193$ $s = 12$ $n = 10$

a) $P\text{-value} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

$= \frac{91.4193 - 100}{12/\sqrt{10}} = -1.3567$

d.f. = 9

$P = 0.025$

b) $\alpha = 5\% = 0.05$

$\therefore$ ปฏิเสธ H_0 (A)

c) $\alpha = 1\% = 0.01$

3. $\therefore$ ปฏิเสธ H_0 (B)

O_i	E_i
310	350
265	250
115	100
310	300

(หน่วย พันบาท)

$\chi^2 = \frac{(O_i - E_i)^2}{E_i}$

$\chi^2 = \frac{(310-350)^2}{350} + \frac{(265-250)^2}{250} + \frac{(115-100)^2}{100} + \frac{(310-300)^2}{300}$
 $= \frac{1600}{350} + \frac{225}{250} + \frac{225}{100} + \frac{100}{300} = 8.05476$

chi table $\rightarrow 0.05 > \chi^2 > 0.025$

d.f. = $(4-1) = 3$
 $\chi^2 = 8.05476$

$\alpha = 0.05 \rightarrow$ ปฏิเสธ

$0.01 \rightarrow$ ไม่ปฏิเสธ

4. $\bar{x} = 80$ $s = 20$ $n = 100$

95% : $80 \pm 1.96 \cdot \frac{20}{\sqrt{100}} = (76.0800, 83.9200)$

97% : $80 \pm 2.17 \cdot \frac{20}{\sqrt{100}} = (75.6600, 84.3400)$

5. $\mu = 64.4$

b) $r = -0.9$